<Kiyuran Naidoo>

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<Personal Statement / Summary>

I am a final-year BSc(Eng) Electrical and Computer Engineering student at the University of Cape Town, driven by a passion for problem-solving and developing innovative, end-to-end solutions to complex, real-world technical challenges. My fascination with the intersection of hardware and software, particularly within embedded systems, artificial intelligence, machine learning, and the 4th Industrial Revolution, fuels my ambition to advance society through technology. I thrive on adaptability, viewing it as both a catalyst and a necessity for the technological advancements across the various industries that are shaping our world.

My academic journey has been enriched by practical programming skills, a strong foundation in physics and mathematics, and a commitment to applying these skills to solve complex problems. Additionally, leadership roles in an elite academic environment and participation in the Toastmasters course have refined my communication, teamwork, and interpersonal skills, which enable me to effectively engage with diverse groups and present ideas with confidence and clarity.

Recent professional experiences spanning full-stack AI platform development, microcontroller architectures, and computer vision have equipped me with the practical skills and adaptability necessary to thrive in fast-paced professional environments. I am eager to bring this blend of technical proficiency, academic rigour, and a collaborative mindset to dynamic vacation work and graduate programmes focused on cutting-edge innovation.

<Education & Achievements>

University of Cape Town | 2023 - 2026

- **Degree**: Enrolled for BSc(Eng) Electrical and Computer Engineering (4-year degree).
- **Academic Honours**: Dean’s Merit List (2023, 2024, 2025).
- **UCT Formula Student Africa (FSA) Racing Team** (Dec 2025 - Present): Currently a member of the Electrical Engineering team.
- **Clubs and Societies:** Active member of the Developer’s Society, AI Society, Engineers Without Borders (EWB), IEEE Society, Consulting Club, Volleyball Club, and Squash Club.

Schneider Electric Bursary | 2025 - 2026

- Awarded for the 2025 and 2026 academic years by Schneider Electric South Africa to support educational pursuits toward my BSc(Eng) Electrical and Computer Engineering degree at UCT.

Glenwood High School Elite Academy | 2018 - 2022

- **Curriculum**: Cambridge Assessment International Education (CAIE) from 2019-2022.
- **Levels**: IGCSE (Equivalent to Grade 10 and 11), AS Level (Equivalent to Matric), and A Level (Equivalent to first-year University).
- **Awards**: Cambridge Academic Colours (Grade 11) and Honours (Grade 12); CAIE Academic Excellence Awards and Distinctions for Mathematics, Computer Science, Physics and Chemistry.
- **Toastmasters**: Youth Leadership Program Certification (2021).

<Professional Experience>

Maxwell+Spark (Pty) Ltd | *Electrical/Software Engineering Internship* | Dec 2025 - Feb 2026

- Architected and built a full-stack web application using React, ASP.NET Core, and Python to facilitate custom dataset capture and train YOLO object detection models.
- Implemented SignalR for real-time WebSocket bi-directional streaming and integrated onnxruntime-web to run live quantised model inference directly in the browser.
- Researched the STM32N6 microcontroller featuring a Neural-ART Accelerator NPU, developed a basic Operating System shell, and implemented an asynchronous, interrupt-driven Command Line Interface (CLI) in C/C++.
- Assembled and soldered 48V PCBs, validated relay switching logic, and constructed a custom In-System Programming (ISP) setup to burn bootloaders onto ATMEGA348P-AU microcontrollers.

AQUA Manufacturing | *Software/Vision Engineering Vacation Work* | Jun 2025 - Aug 2025

- Contributed to a Masters-level research initiative by master's student Timothy Reddy at UCT, focusing on an "Adaptive Quality Upscaling with Advanced Manufacturing" system.
- Developed Python scripts utilising OpenCV to capture images from a Raspberry Pi Camera 3 and applied initial filtering to isolate laser lines for a Laser Triangulation 3D-Scanning System.
- Implemented mathematical frameworks for 3D triangulation to convert 2D laser coordinates into spatial points, and developed routines to export the data into standard point cloud formats (.STL and PLY).

<Skills, Proficiencies & Interests>

Software & Programming Languages

- **High-Level & Scripting:** Python, Java, MATLAB.
- **Low-Level & Hardware Description:** C, C++, Verilog.
- **Web & Database:** React, ASP.NET Core, SQL.

Hardware & Embedded Systems

- **Microcontrollers:** STM32 architectures (such as STM32N6), ATMEGA series (such as ATMEGA328P), Arduino platforms.
- **Communication Protocols:** SPI, I2C, UART, WebSockets (SignalR), REST.
- **Hardware Implementation:** In-System Programming (ISP), bootloader flashing, relay switching logic, PCB assembly.

Machine Learning & Computer Vision

- **AI/ML Frameworks:** OpenCV, ONNX Runtime.
- **Applied AI:** Edge AI inference, YOLO object detection, Convolutional Neural Networks (CNNs), image classification, feature extraction.
- **Vision Systems:** Laser triangulation 3D-scanning pipelines, real-time video streaming, point cloud generation.

Engineering Tools & DevOps

- **PCB & CAD Design:** KiCad, LTSpice, SolidWorks.
- **Development Environments & Version Control:** STM32CubeIDE, Docker, Git.

Domain Expertise (gained from work experience) & Core Competencies

- **Specialised Knowledge:** Battery Management Systems (BMS), lithium-ion systems, renewable energy technologies.
- **Systems Engineering:** Full-stack software architecture, asynchronous/interrupt-driven state machine design, circuit design and prototyping.
- **Research & Documentation:** Academic and professional technical reporting, stringent citation formatting (i.e. IEEE).

Interests & Hobbies:

- Technology & AI, Astronomy, Music Production, Art, Outdoors & Hiking, Travel, Sport, Gym & Exercise.